



CONFIDENTIAL — FOR INTERNAL USE

Smart Proctored Examination System

Executive Presentation Guide

A comprehensive overview of our intelligent online examination platform — designed for institutional leadership, academic deans, examination controllers, and administrative authorities.

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1. The Problem We Are Solving

Traditional online examination methods have a fundamental weakness: **the institution has no control over what a student does on their personal device during a test.**

Without Our System

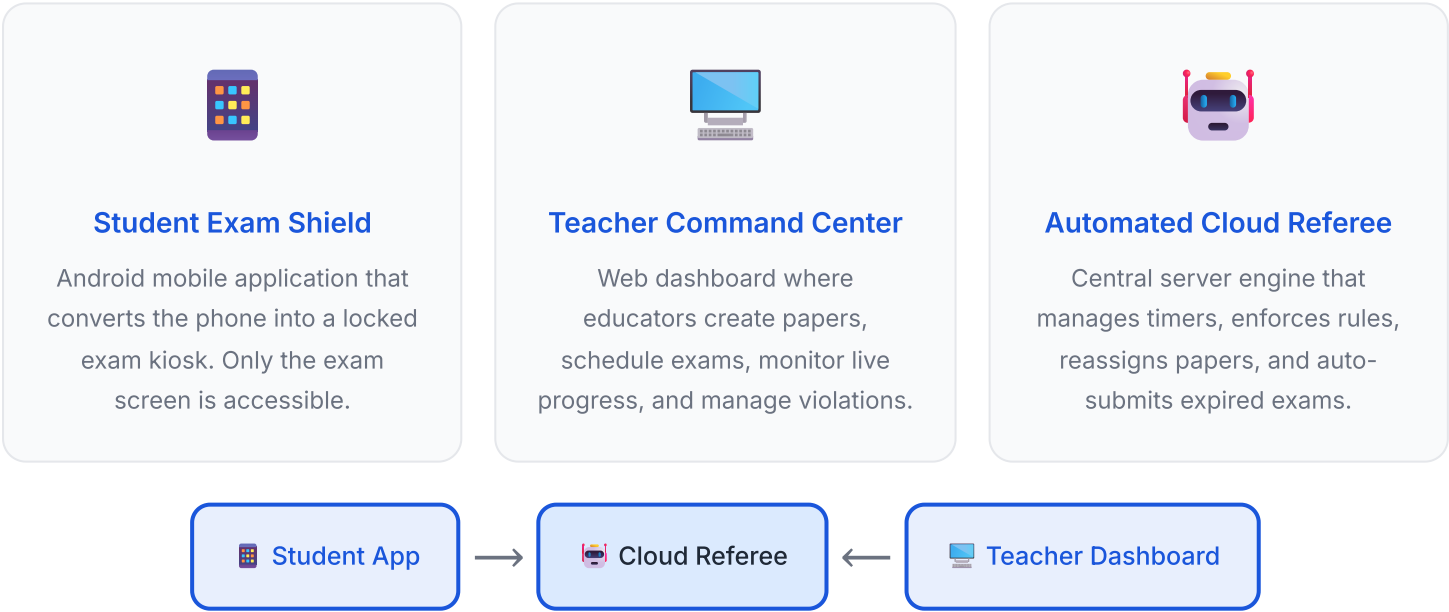
- Student minimizes exam to search Google
- Screenshots shared on WhatsApp groups
- Disconnecting internet "pauses" the clock
- Same questions given if student retries
- No live supervision — cheating found too late
- Hours of manual MCQ paper grading

With Our System

- Phone becomes a single-purpose exam device
- Screenshots produce completely black images
- Server controls timer — disconnect does not help
- Brand new question set assigned every retry
- Faculty watches live with instant intervention
- Automatic instant MCQ grading on submission

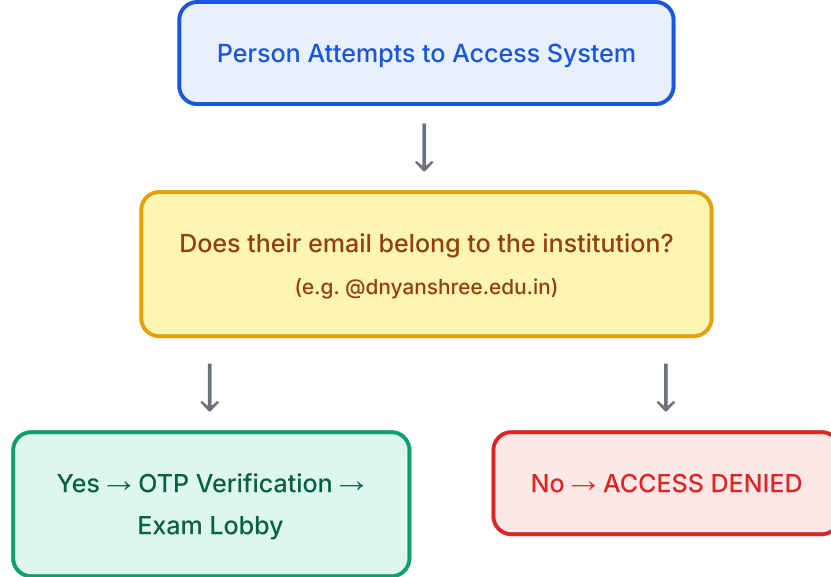
2. System Architecture: Three Connected Pillars

Our platform combines three intuitive tools working together in complete synergy:



3. Who Can Access the System?

Security begins well before a single question is displayed. Our platform includes multiple verification layers:

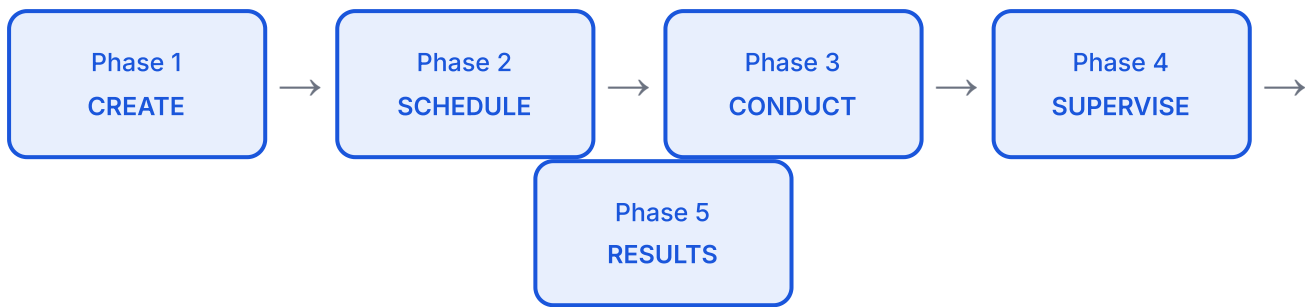


Key Access Controls:

- **Institutional Email Whitelisting:** Only accounts under the institution's official email domain are permitted. Personal Gmail, Yahoo, or Outlook accounts are automatically rejected.
- **Phone OTP Verification:** Each student profile is linked to a verified mobile number using one-time passwords, preventing impersonation.
- **Device Administrator Permission:** Students must grant the app special device privileges to disable the camera, prevent screenshots, and enforce lockdown. Without consent, the exam cannot begin.

4. Exam Lifecycle: From Creation to Results

The complete journey of an examination follows five clear phases:



Phase 1: Create — Question Paper Preparation

- Faculty logs into the Teacher Command Center from any computer or laptop.
- Creates a new exam paper by entering: **Title**, **Subject**, and **Duration** (in minutes).
- Adds MCQ questions one by one, each with four options and the correct answer marked.
- Questions can include text, mathematical formulas, circuit diagrams, or medical images.
- All uploaded images are **automatically compressed** to ensure fast loading on slower networks.

Phase 2: Schedule — Timetable Configuration

- Faculty sets the **Opening Date and Time** (e.g., Monday, 10:00 AM) and **Closing Date and Time** (e.g., Monday, 11:30 AM).
- Students attempting to start outside this window are automatically blocked by the server.
- Multiple papers can be created for the same subject (used for automatic reassignment if a student cheats).

Phase 3: Conduct — Student Takes the Exam

- Student opens the app, signs in with institutional credentials, and enters the exam lobby.
- Upon starting, the phone enters **lockdown mode**: camera disabled, screenshots blocked, AI assistants blocked.
- Questions appear one at a time. After answering, the student taps "Next" and **cannot return** to previous questions.
- A visible countdown timer shows time remaining.

Phase 4: Supervise — Faculty Monitors Live

- Faculty sees a live table of every student currently taking the exam with color-coded status badges.
- If a violation occurs, the student's row highlights immediately with warning indicators.
- Faculty can click to review details and take action (override or malpractice) instantly.

Phase 5: Results — Automatic Scoring and Reports

- All answers are scored instantly upon submission against the correct answer key.

- Dashboard displays score distributions, batch averages, and violation history as visual charts.
- Complete records can be downloaded as **CSV/Excel spreadsheets** for institutional archives.

5. Intelligent Anti-Cheat: Fair to Accidents, Strict on Cheating

How does the system distinguish between a genuine emergency and a deliberate cheating attempt? It reads real-time hardware signals from the phone's operating system automatically:

SAFE — Zero Penalty

- **Incoming Phone Call** — Timer pauses. Student resumes after the call.
- **Screen Lock / Sleep** — Timer pauses. Student resumes after unlocking.

No warning recorded. No penalty. Completely safe.

VIOLATION — Exam Terminated

- **Home Button** — Leaves exam to home screen.
- **App Switch** — Opens browser, calculator, or messaging.
- **Notification Shade** — Pulls down notification panel.

Exam closes immediately. Warning strike recorded.

Note

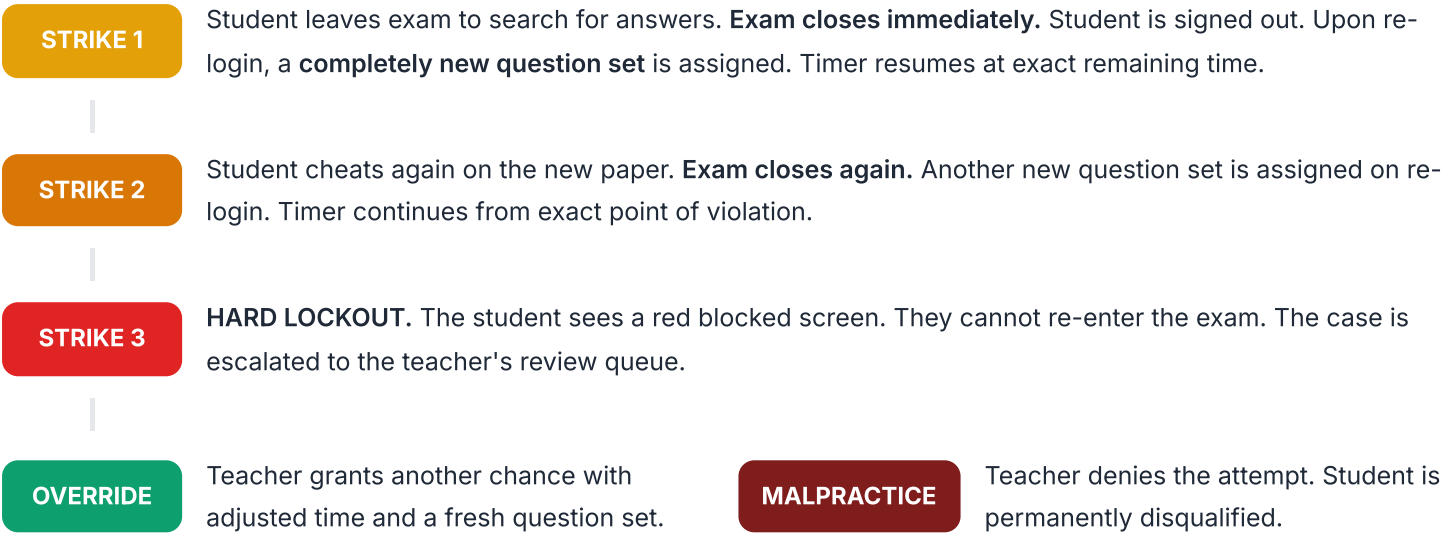
These distinctions are made automatically using hardware-level signals. No manual configuration is needed by faculty, and no student input is required.

Key Security Features

Feature	How It Works
Screenshot and Recording Proof	Any capture attempt produces a completely black image. Questions cannot be photographed or shared digitally.
Camera and AI Lockdown	Phone camera hardware and AI overlays (Google Lens, Circle-to-Search) are disabled during the exam.
One-Way Question Flow	Once answered, students cannot go back. This prevents group cheating where later questions provide contextual clues.
Interactive Diagram Zoom	Students can pinch-to-zoom on circuit diagrams or medical images without leaving the secure exam screen.

6. The Three-Strike Rule: Managing Cheating

When a student commits a violation, the system follows a structured, transparent, and fair escalation process:



Three Key Principles

- 1. **Fresh Questions Every Time** — Cheating is futile because the student never sees the same questions again.
- 2. **Exact Time Preservation** — If 4:30 remained at cheating, the new paper starts at exactly 4:30. No time gained or lost.
- 3. **Faculty Has Final Authority** — The system locks the student, but the teacher decides whether to forgive or fail.

7. What Faculty and Administration Can See

Live Monitoring Dashboard

Information	Description
Student Name and ID	Which student is currently in the exam
Current Question	How far the student has progressed (e.g., Question 7 of 25)
Time Remaining	Live countdown showing exact seconds left
Status Badge	Color-coded: Active (green), Paused (yellow), Warning (orange), Blocked (red)
Violation Count	Number of warnings accumulated (0, 1, 2, or 3)
Quick Actions	One-click buttons: Review, Override, or Deny

Analytics and Reports

Feature	What It Shows
Score Distribution Chart	Bar graph showing how many students scored in each percentage range
Violation Frequency Graph	Trend line showing cheating attempt patterns across sessions
Batch Summary	Average scores, highest/lowest marks, completion rates
Individual Student Cards	Per-student breakdown: answers, time per question, violations
CSV/Excel Export	One-click download of complete exam data for institutional records

8. Institutional Benefits

Advantage	Explanation
Precise Scheduling	Set exact opening and closing times. The system enforces them automatically — no early birds, no latecomers.
Automatic Collection	When time expires, the exam auto-submits instantly. No "my internet was slow" excuses.
Bandwidth Friendly	All images are auto-compressed before delivery. Works on slow mobile data and congested campus Wi-Fi.

Real-Time Updates	Dashboard updates instantly without page refreshes. Faculty sees activity the moment it happens.
Instant Results	MCQ scoring happens automatically upon submission. No manual paper checking required.
Audit-Ready Archives	Complete exam history stored and downloadable, supporting NAAC, NBA, and internal audits.
No Special Hardware	Students use existing Android smartphones. No tablets, laptops, or lab computers needed.
Scalable	Cloud referee handles hundreds of simultaneous exam-takers without degradation.

9. Real-World Exam Day Walkthrough

Here is a typical examination day from start to finish:

8:30 AM	●	Prof. Sharma logs into the Command Center from their office laptop. Reviews the 25-question MCQ paper on Basic Electronics.
9:00 AM	●	Prof. Sharma publishes the paper and sets the exam window: 10:00 AM to 11:00 AM with a 45-minute duration.
9:45 AM	●	Students open the app. The lobby shows "Basic Electronics — Starts at 10:00 AM". The Start button is greyed out.
10:00 AM	●	Exam opens automatically. Students begin. Phones enter lockdown. The 45-minute server countdown begins.
10:15 AM	●	Student Ravi presses Home to open Google. System instantly detects unauthorized exit. Ravi's exam closes. Strike 1.
10:16 AM	●	Ravi logs back in. Server assigns a completely different question set (Paper B). Timer resumes at 30 minutes remaining.
10:35 AM	●	Student Priya receives a phone call from her parent. Timer pauses automatically. No penalty. She finishes the call and resumes.
10:45 AM	●	Most students finish and tap Submit. Scores calculated instantly.
11:00 AM	●	Exam window closes. Any remaining active exams are auto-collected by the server.
11:01 AM	●	Prof. Sharma opens the Analytics tab. Score charts are already generated. She downloads the CSV report for records.

10. Frequently Asked Questions

Q: Do students need expensive phones?

No. Any standard Android smartphone (Android 8.0 or newer) works. No special hardware is required.

Q: What if a student's phone battery dies mid-exam?

The server preserves their exact progress and remaining time. When the student charges and logs back in, they resume from where they left off with the same timer.

Q: Can students cheat by using two phones?

Each student account can only have one active session. Starting on a second device terminates the first. Screenshots are blocked, so there is nothing to photograph from a second device's camera.

Q: What if campus Wi-Fi goes down during an exam?

The server-side timer does not penalize network disconnections. Students reconnect and resume. All previously submitted answers are already saved on the server.

Q: Can the exam be taken on laptops or desktops?

The student app is Android-only by design. Web browsers cannot enforce the hardware-level security locks (camera disabling, screenshot blocking, app-switching prevention) that our anti-cheat model requires.

Q: How many students can take the exam simultaneously?

The cloud infrastructure scales automatically. Hundreds of concurrent students are supported without performance issues.

Q: Is student data secure?

Yes. All communication is encrypted. Student answers are stored in secured cloud databases with role-based access controls. Only authorized faculty can view results.

Q: Can multiple departments use this simultaneously?

Absolutely. Each paper is tagged with a subject and can be independently scheduled. Multiple departments can run exams simultaneously without any interference.

11. Future Roadmap

Planned Enhancement	Description
Push Notifications	Students receive instant mobile alerts when exam schedules are published or when a teacher grants an override.
Parent/Guardian Portal	A read-only view where parents can see their ward's exam results and attendance records.
Multi-Language Support	Question papers displayed in regional languages (Marathi, Hindi) alongside English.

Offline Exam Mode	For locations with unreliable internet, answers cached locally and synced when connectivity returns.
Advanced Analytics	Question-level difficulty analysis, topic-wise performance breakdowns, and semester-over-semester trends.